

SRINATH YADAV BOLLEBOINA

Machine Learning Engineer

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SUMMARY

Machine Learning Engineer with 3+ years of experience developing and deploying machine learning models within Agile and Waterfall SDLC frameworks across the project lifecycle. Expertise in Python, and C++ programming, with deep learning applications in CNNs, RNNs, and NLP tools like BERT and GPT-3. Proficient in big data processing with Apache Spark and Hadoop and model deployment on AWS, Azure, and GCP. Skilled in data visualization using Tableau and Power BI to communicate insights and drive informed decision-making in complex environments.

SKILLS

Methodology: SDLC, Agile, Waterfall
Programming Language: C, C++, Python, SQL, Java, Kotlin, HTML, CSS, PHP, React, Angular, Node.js
IDE's: PyCharm, Jupyter Notebook, Visual Studio Code
Machine Learning Frameworks and Libraries: TensorFlow, Keras, PyTorch, Scikit-learn, XGBoost, LightGBM
Data Processing and Analysis: NumPy, Pandas, SciPy, Matplotlib, Seaborn
Deep Learning: Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Long Short-Term Memory Networks (LSTM), Generative Adversarial Networks (GAN)
Natural Language Processing (NLP): NLTK, SpaCy, BERT, GPT-3 Transformer models
Big Data Technologies: Apache Hadoop, Apache Spark, Apache Kafka
Databases: MySQL, PostgreSQL, MongoDB
Cloud Platforms: Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform (GCP)
Development Tools: Jupyter Notebooks, Git, Docker, Kubernetes, Apache Airflow
Model Deployment: Flask, Fast API, TensorFlow Serving, MLflow
Data Visualization: Tableau, Power BI, Plotly
Mathematics and Statistics: Linear Algebra, Calculus, Probability and Statistics
Other Skills: Feature Engineering, Model Selection and Evaluation, Hyperparameter Tuning, Cross-Validation A/B Testing, Data Cleaning and Preprocessing, Critical Thinking, Communication Skills, Team Collaboration
Operating Systems: Windows, Linux

EDUCATION

Masters in Computer Science University of Alabama at Birmingham, Birmingham, AL	April 2024
Bachelor in Computer Engineering Vidya Jyothi Institute of Technology, Hyderabad, India	July 2020

EXPERIENCE

ServiceNow, AL Machine Learning Engineer	Nov 2023 – Current
• Designs and deploys end-to-end machine learning solutions within Agile frameworks, delivering production-ready models that enhance business processes. • Develops predictive models using TensorFlow, Keras, and Scikit-learn, driving measurable improvements in forecasting accuracy for multiple business units. • Implements data processing pipelines with Apache Spark and Hadoop, ensuring efficient data ingestion and transformation for model training and evaluation. • Conducts NLP projects using BERT and GPT-3 to derive insights from unstructured text data, enabling targeted customer experience strategies. • Builds interactive dashboards in Tableau and Power BI, empowering stakeholders with actionable insights and reducing report generation time by 30%. • Utilizes Docker and Kubernetes to containerize and deploy models, maintaining scalability and reducing deployment time. • Optimizes feature engineering processes and hyperparameter tuning to maximize model performance across diverse datasets.	
Capgemini Data Scientist	Jan 2020 – Dec 2022
• Collaborated with cross-functional teams within SDLC and waterfall frameworks, aligning data science initiatives with business goals and optimizing project delivery timelines. • Developed and deployed machine learning models using TensorFlow and PyTorch, which enhanced predictive accuracy across customer behavior analysis projects, resulting in more targeted marketing strategies. • Designed data preprocessing workflows with Apache Spark and Hadoop, significantly reducing data pipeline execution time and improving data quality for model training. • Built NLP models using BERT and GPT-3 for text classification and sentiment analysis, enabling more precise customer feedback processing and enhancing product insights. • Conducted feature engineering and hyperparameter tuning, optimizing model performance and delivering a 15% improvement in model efficiency across various business units. • Led A/B testing and cross-validation studies, enabling better assessment of model outcomes and ensuring reliable predictions within critical applications. • Automated ETL processes with Apache Airflow, streamlining data flows and reducing manual intervention in data management tasks by approximately 40%. • Utilized Tableau and Power BI to create interactive visualizations, empowering stakeholders with clear, actionable insights that drove strategic decision-making across teams.	

PROJECTS

File Sharing Web Application | UAB

- Designed and developed a Flask-based web application for secure file sharing.
- Integrated AWS S3 and SNS for scalable storage and email notifications
- **Technologies:** Python, Docker, Flask, AWS S3, AWS SNS, PostgreSQL, HTML/CSS.

Demonstrates expertise in web development and cloud integration.

Invest Match Web Application | UAB

- Designed and developed an API-based platform connecting investors and companies.
- Utilized Fast API for the server side and React JS for seamless data flow integration with the database.
- **Technologies:** React JS, Fast API (server-side), Bootstrap, and MongoDB (database).

Demonstrates expertise in frontend and API integrations with database

Disaster Tweet Prediction | UAB

- Performed sentiment analysis for the tweets to classify them as disaster or non-disaster using Deep Learning models.
- Recurrent Neural Networks (RNNs) with Long Short-Term Memory (LSTM) and Convolutional Neural Networks (CNNs) were used to compare the predictions of both models.
- Implemented a Self-coded version for both models and compared the accuracy and validation loss.

Demonstrates expertise in sentiment analysis and disaster prediction using Deep Learning models, with hands-on experience in implementing and comparing RNN, LSTM, and CNN architectures.

PDF QuizBot | Personal Project

- Designed and built an intelligent web application that auto-generates quizzes from uploaded PDF files using Streamlit and Langchain, seamlessly integrating into Google Cloud's Vertex AI.
- **Core Features:** Real-time quiz creation, interactive Q&A with PDF content, and advanced NLP for precise content extraction.
- Developed an innovative "Chat with PDF" feature, allowing users to ask natural language questions directly to PDF documents and receive relevant, contextual responses.
- **Technologies:** Python, Streamlit, Langchain, Google Cloud Vertex AI, ChromaDB, Firebase

Demonstrates expertise in AI-driven application development, real-time data processing, and innovative feature integration.

Certification

MuleSoft Certified Developer Level -1